Adrian Matei

Backend Engineer – Node.js / TypeScript | Production Debugging & Distributed Systems | Brasov, Romania | Willing to relocate to Bucharest

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Summary

Backend engineer with 5+ years in hands-on engineering roles, including 4 years of TypeScript and 2+ years building Node.js/NestJS services. Most recently owned features for a global enterprise SaaS platform from code through CI/CD into production, monitoring and debugging them in Splunk. Detected and remediated a live production server compromise as sole engineer for a startup, and debugged card-payment processing (preauthorization, card/EBT) by tracing transaction data across layers. Builds and instruments distributed systems with Kafka, RabbitMQ, Kubernetes and OpenTelemetry, and builds with Claude Code and Cursor.

Skills

Core: Node.js, TypeScript, JavaScript, NestJS

Observability & Debugging: Splunk, OpenTelemetry (distributed tracing), Prometheus, Grafana, Loki, Tempo, structured logging, health checks

Microservices & Integrations: microservices, API gateway, REST, OpenAPI/Swagger, event-driven architecture, Kafka (consumer groups), RabbitMQ (retries, dead-letter queues), third-party API integrations (payment, booking, Jira, Microsoft Graph)

Databases: SQL: PostgreSQL, MySQL; NoSQL: MongoDB (aggregation pipelines), Redis, DynamoDB; TypeORM, Mongoose, Prisma

Cloud & Infrastructure: Kubernetes (k3s), Helm, Docker, AWS, Terraform, Ansible, Linux server hardening (SSH, firewall)

CI/CD & Testing: GitHub Actions, Jenkins, Spinnaker, Git, Jest, Vitest, Supertest (unit + e2e), Playwright

AI coding & agent tools: Claude Code (plan-then-implement agent workflow), Cursor; LLM integration via the Claude API

Auth & Security: JWT, OAuth 2.0, session auth (Passport), RBAC, OWASP Top 10

Also worked with: Python, Go, C#/.NET (ASP.NET Core), Java, Angular, React

Work Experience

Senior Software Engineer, Brainspotting | Technology Recruitment – 4-month contract, Remote Jan 2026 – Apr 2026

- Contracted to a global enterprise SaaS company, working in a distributed international team on a multi-service NestJS monorepo
- Owned features end to end through CI/CD (Jenkins, Spinnaker) into production, monitoring and debugging production behaviour in Splunk
- Wrote Axios-based adapter layers for upstream services with standardised error handling, normalising third-party failures into consistent application-level errors
- Built and extended REST endpoints against OpenAPI specs (paginated list endpoints, nested subresources and aggregation endpoints), with class-validator DTOs enforcing request contracts
- Implemented DynamoDB-backed repository layers on AWS SDK v3, designing keyed access patterns and transactional writes behind a persistence interface the service layer consumed
- Wrote unit and e2e tests; introduced E2E coverage reporting, identifying gaps between total and end-to-end coverage
- Used Cursor to navigate the multi-service architecture and deployment patterns and speed up testing and documentation

Full-Stack Software Engineer, Ruuts.travel – Part-time, Remote Mar 2025 – Dec 2025

- Sole engineer for an early-stage travel-booking startup, owning the full stack and production infrastructure end to end
- Detected and fully remediated a live server compromise: traced unauthorized cron jobs opening outbound connections to hundreds of external servers, removed malicious processes and persistence mechanisms, hardened SSH and firewall access, and blocked the malicious traffic

- Audited the payment integration after signs of tampering and restored its integrity
- Delivered features in an ASP.NET Core/Angular monorepo, integrating third-party booking and payment APIs

Software Engineer, Ensemble Software – Remote Nov 2022 – Sep 2025

- Debugged Java POS payment processing (card/EBT transactions, multiple entry modes, preauthorization flows), tracing transaction data across application layers
- Proactively resolved critical production issues when identified, minimizing downtime and user disruption
- Led the design and development of a NestJS API from scratch for a collaborative requirements-management web app, modelling a PostgreSQL schema for hierarchical, versioned project data
- Implemented the full JWT authentication flow (registration with email confirmation, login, forgot/reset password); containerised services with Docker
- Extended third-party integrations: reworked the Jira export to mirror the requirement tree as epics, stories and sub-tasks, and built a SharePoint integration on Microsoft Graph (OAuth 2.0)
- Introduced a requirements and progress-tracking system that improved client communication, later adopted as the team standard; led the team's client demos

Software Developer, Talenting Software – Hybrid Oct 2021 – Oct 2022

- Built Azure Blob image upload and PDF conversion for a floor-plan review module with version history

Software Developer Intern, Trimble, Siemens, Talenting Software – Remote Nov 2020 – Oct 2021

- Built full-stack web apps with ASP.NET Core and Angular

Projects

DevBoard – event-driven microservices platform (NestJS, Kafka, RabbitMQ, Kubernetes)
github.com/AdrianMatei20/devboard

- Instrumented all services with OpenTelemetry via a Collector to Tempo, Prometheus and Loki, giving one connected trace from the gateway through downstream services into Kafka
- Designed the async integration layer: services communicate only through Kafka (3-partition topic, consumer groups fanning out to notification, analytics and audit) and RabbitMQ (webhooks with exponential backoff over 3 attempts and a dead-letter queue); event publishing is best-effort, so a broker outage degrades the pipeline without failing the write
- Built the NestJS API gateway as the single entry point: strips client-supplied identity headers, rejects malformed requests before any JWT crypto runs, validates the token and injects identity headers downstream services trust
- Deployed on Kubernetes (k3s) via Helm charts with liveness/readiness probes and ConfigMap/Secret separation, provisioned with Terraform and Ansible; GitHub Actions pipelines test, build, push SHA-pinned images and verify the rollout
- Built MongoDB-backed services (Mongoose profile service, Go analytics service with aggregation pipelines) alongside NestJS

QuizForge – LLM quiz generation (NestJS, PostgreSQL, Redis, Claude API) github.com/AdrianMatei20/QuizForge

- Built a resilient third-party (LLM) integration layer: daily budget guard, Redis cache, retries with jittered backoff, and zod schema validation with one repair call
- Streamed results as JSON Lines over Server-Sent Events: each item is validated and saved as soon as its line completes, and reconnecting clients replay saved items before live events
- Tested with Vitest unit and Supertest e2e suites on real PostgreSQL and Redis; GitHub Actions also runs a Playwright test on the full Docker stack and fails the build on OpenAPI or generated-type drift
- Built with Claude Code in 11 phases, approving a written plan for each and verifying it against lint and tests

Taskflow – project management API (NestJS, PostgreSQL, Redis, AWS) github.com/AdrianMatei20/NestJsAPI

- Persisted structured logs to PostgreSQL, queryable by level and context, and exposed health checks plus request-rate and resource metrics to Prometheus and Grafana
- Added Redis caching with TTL control and graceful fallback on cache failure; covered REST and GraphQL APIs with Jest unit and e2e tests and automated deployment to AWS EC2 via GitHub Actions

Education

Transilvania University of Brasov – MSc in Modern Technologies in Software Systems Engineering — Jun 2023

Transilvania University of Brasov – BSc in Computer Science — Jun 2021

Certificates

AWS Certified Solutions Architect – Associate — In preparation

Cambridge English: Advanced (CAE) - C1 — Jun 2017

YouniHack 2025 – 24-hour cybersecurity hackathon (OWASP Top 10, CTF-style challenges) — Sep 2025